EXCEL LEARNING HANDBOOK

PART 3 | LESSONS 12–15

Filtering • Sorting • Custom Sorting • Advanced Filtering

A consolidated reference for the concepts learned and demonstrated in the Excel transcript sessions. Includes commands, reproducible sample datasets, exact filtering criteria, expected results, and key cautions. Practice assignments and quizzes have intentionally been excluded.

Contents

Lesson	Coverage
12 – Filtering	Filter setup; text, date, time and number filters; multi-column AND filtering; search; clear and remove filters
13 – Sorting	Dates, text and numbers; ascending/descending; sorting filtered records; avoiding misaligned rows
14 – Custom Sorting	Sort dialog; Add Level; first and second priority; agent and contact time; two-/three-level sorting
15 – Advanced Filtering	Criteria ranges; AND/OR; comparison operators; exact matches; in-place and copy output; cross-sheet extracts; unique records

Workbook conventions

Illustrations use the same training workbook Excel_Complete_Practice.xlsx and sheets 12_Filtering, 13_Sorting, 14_CustomSorting, and 15_AdvancedFiltering. Dates in these examples refer to 2023 sample records, not today’s date. Excel menu labels may vary slightly between versions.

LESSON 12 | Filtering in Excel

12.1 Purpose and behaviour

Filtering is a data-interpretation tool that shows only matching rows while temporarily hiding the others. The source records are not deleted, and visible row numbers can skip. In the pizza-shop call scenario, the goal is to examine Query, Complaint or Request calls, their timings, and operational performance.

12.2 Dataset: pizza-shop customer contacts

ID	Date	Time	Agent	Type	Duration	Answers	Product	Rating
C101	02-Jan-23	10:30 AM	Mike Ross	Query	180	45	Crispy Pizza	8
C102	05-Jan-23	1:15 PM	Harvey	Complaint	260	90	Cheese Pizza	5
C103	07-Jan-23	11:45 AM	Rachel Zane	Request	120	35	Crispy Burger	9
C104	10-Jan-23	2:30 PM	Mike Ross	Query	200	65	Crispy Burger	7
C105	12-Jan-23	9:20 AM	Harvey	Complaint	310	110	Cheese Pizza	4
C106	15-Jan-23	12:45 PM	Mike Ross	Query	150	55	Crispy Pizza	8
C107	18-Jan-23	3:10 PM	Rachel Zane	Request	90	40	Veg Burger	9
C108	20-Jan-23	11:30 AM	Harvey	Complaint	280	85	Crispy Pizza	6
C109	24-Jan-23	5:00 PM	Mike Ross	Query	175	70	Veg Pizza	8
C110	28-Jan-23	10:15 AM	Mike Ross	Request	140	30	Crispy Burger	10

> Important: The lesson’s earlier worksheet had Agent Handled in column D, Contact Type in E, Contact Date in B, Contact Time in C, Time to Answer in G, and Product Name in H. The compact display above reorders columns for readability; use your actual header names when filtering.

12.3 Turn filters on

1. Click any cell within the connected dataset and press Ctrl+A (or explicitly select the entire table, including header).

2. Choose Home → Sort & Filter → Filter, or press Ctrl+Shift+L.

3. Observe one dropdown arrow on each column header; click a dropdown to choose the desired values.

A header row is important. If headers start at row 3, select the correct range beginning at row 3. Selecting blank/title rows accidentally may cause filter arrows on the wrong row.

12.4 Text checklist filter: Contact Type = Query

1. Open the Contact Type dropdown.

2. Uncheck Select All, check only Query, then click OK.

Matching contact IDs	Count
C101, C104, C106, C109	4

The dropdown lists distinct values (e.g., Complaint, Query, Request), rather than a separate option for every row.

12.5 Agent filter: Mike Ross

1. Clear the Contact Type condition using Clear Filter From Contact Type.

2. Open Agent Handled and choose only Mike Ross.

Matching IDs	Count
C101, C104, C106, C109, C110	5

12.6 Date filter: before 7 January 2023

1. Clear prior conditions using Data → Clear.

2. Open Contact Date → Date Filters → Before.

3. Enter 07-Jan-2023; select OK.

Matching IDs	Reason
C101 and C102	02-Jan and 05-Jan are strictly before 07-Jan

> Important: “Before” excludes the date itself. Excel must store the source values as real dates rather than text. Use an unambiguous date format to avoid locale confusion.

12.7 Date filter: between 7 and 12 January 2023

1. Choose Contact Date → Date Filters → Between.

2. Enter lower boundary 07-Jan-2023 and upper boundary 12-Jan-2023; confirm.

Matching IDs	Date
C103	07-Jan-2023
C104	10-Jan-2023
C105	12-Jan-2023

The Between selection is inclusive: both endpoints qualify. This differs from combining strictly After and strictly Before.

12.8 Time filter: 10 AM through noon

1. Choose Contact Time dropdown → Number Filters → Between (menu wording can vary by Excel version).

2. Enter 10:00 AM and 12:00 PM.

Matching IDs	Time
C101	10:30 AM
C103	11:45 AM
C108	11:30 AM
C110	10:15 AM

Excel stores a time as part of a day: 06:00 AM = 0.25, 12:00 PM = 0.5, and 06:00 PM = 0.75. These are numeric values, so the transcript's workbook offered Number Filters for time.

12.9 Combined filters across four fields

Business question: Show records with Contact Date after 12-Jan-2023, Contact Time from 11:30 AM through 3:00 PM, Time to Answer <100 seconds, and Product Name containing "Crispy."

1. Clear previous criteria and ensure filters are enabled.

2. Contact Date → Date Filters → After → 12-Jan-2023.

3. Contact Time → Number Filters → Between → 11:30 AM to 3:00 PM.

4. Time to Answer → Number Filters → Less Than → 100.

5. Product Name → Text Filters → Contains → Crispy (or use the dropdown Search box).

Matching ID	Date	Time	Answer time	Product
C106	15-Jan-23	12:45 PM	55	Crispy Pizza

Filters on different columns are combined with AND logic: each visible record must pass all four conditions. To check the result, clear one condition at a time and observe the rows that reappear.

12.10 Search, clear, and remove filters

Action	Result
Dropdown search: type Crispy	Finds matching product options in the list
Data → Clear	Removes active restrictions; filter arrows remain
Clear Filter From [Column]	Clears just one column's restriction
Ctrl+Shift+L or Home → Filter	Turns AutoFilter on/off; turning it off removes arrows

LESSON 13 | Sorting in Excel

13.1 What sorting does

Sorting rearranges the order of records. Unlike filtering, it does not hide rows just because they fail a condition. The full associated row should travel together so dates, names and amounts remain matched.

Type	Ascending	Descending
Text	A to Z	Z to A
Number	Smallest to Largest	Largest to Smallest
Date	Oldest to Newest	Newest to Oldest
Time	Earliest to latest	Latest to earliest

13.2 Dataset for basic sorting

Contact ID	Contact Date	Agent	Time	Rating
C101	10-Jan-23	Mike Ross	2:30 PM	8
C102	02-Jan-23	Harvey	11:15 AM	5
C103	15-Jan-23	Rachel Zane	4:45 PM	9
C104	05-Jan-23	Mike Ross	10:00 AM	7
C105	01-Jan-23	Harvey	9:30 AM	4
C106	12-Jan-23	Rachel Zane	1:20 PM	8
C107	07-Jan-23	Mike Ross	11:45 AM	6
C108	03-Jan-23	Harvey	3:10 PM	7

13.3 Oldest to Newest / Newest to Oldest

1. Select a date value in Contact Date, rather than the header text.

2. Home → Sort & Filter → Sort Oldest to Newest (or Data tab sort controls).

3. Observe complete rows moving with their Contact IDs and Agent names.

Oldest → newest IDs	Newest → oldest IDs
C105, C102, C108, C104, C107, C101, C106, C103	C103, C106, C101, C107, C104, C108, C102, C105

> Important: If Excel displays text-based sorting for a date column or sorts dates strangely, check whether entries were stored as text rather than true date values.

13.4 Sort text and numeric ratings

1. Click a value in Agent Handled → Home → Sort & Filter → Sort A to Z.

2. For ratings, click a numeric cell → Sort Largest to Smallest.

Sort	Expected order
Agents A → Z	Harvey, Harvey, Harvey, Mike, Mike, Mike, Rachel, Rachel
Ratings high → low	9, 8, 8, 7, 7, 6, 5, 4

13.5 Filter a day, then sort its agents

The lecture's second sorting example filters 24-Feb-2023 calls, then groups the visible calls alphabetically by Agent Handled.

ID	Contact Date	Agent	Time
C201	24-Feb-23	Mike Ross	10:30 AM
C202	23-Feb-23	Harvey	11:15 AM
C203	24-Feb-23	Rachel Zane	2:45 PM
C204	24-Feb-23	Harvey	9:30 AM
C205	25-Feb-23	Mike Ross	1:20 PM
C206	24-Feb-23	Mike Ross	4:10 PM
C207	24-Feb-23	Harvey	3:30 PM
C208	23-Feb-23	Rachel Zane	12:15 PM

1. Turn on filters; Contact Date dropdown → choose only 24-Feb-2023.

2. Open Agent Handled dropdown → Sort A to Z (or use Home → Sort & Filter).

Visible ID order after filter + sort	Agents shown
C204, C207, C201, C206, C203	Harvey; Harvey; Mike; Mike; Rachel

13.6 Sorting safety

Sort the whole connected dataset, not a detached salary/date column. If prompted, choose Expand the selection so rows remain consistent. Confirm "My data has headers" in the Sort dialog when applicable. Dates and numbers stored as text can sort incorrectly. Alt → H → S opens Home Sort & Filter through sequential ribbon key presses on Windows.

LESSON 14 | Custom (Multi-Level) Sorting

14.1 Why use a custom sort?

Custom Sort defines priorities explicitly. First arrange records by agent, then sort each agent's calls chronologically. Running separate sorts in an arbitrary order can lose the intended within-group order; a multi-level sort expresses the hierarchy directly.

14.2 Sample call records

ID	Date	Agent	Contact Time	Duration s
C201	24-Feb-23	Mike Ross	4:30 PM	180
C202	23-Feb-23	Harvey	11:15 AM	260
C203	24-Feb-23	Rachel Zane	2:45 PM	120
C204	24-Feb-23	Harvey	3:30 PM	200
C205	25-Feb-23	Mike Ross	1:20 PM	310
C206	24-Feb-23	Mike Ross	10:00 AM	150
C207	24-Feb-23	Harvey	9:30 AM	280
C208	23-Feb-23	Rachel Zane	12:15 PM	175
C209	24-Feb-23	Mike Ross	11:45 AM	140
C210	24-Feb-23	Harvey	1:00 PM	220

14.3 Filter the day and open Custom Sort

1. Filter Contact Date to 24-Feb-2023; seven matching records remain.

2. Click any cell in the source table and select Data → Sort, or Home → Sort & Filter → Custom Sort.

3. Confirm My data has headers is checked.

14.4 Configure two levels: agent first, time second

Priority	Column	Sort On	Order
Sort by	Agent Handled	Cell Values	A to Z
Then by	Contact Time	Cell Values	Smallest to Largest

1. Under Sort by, choose Agent Handled and A to Z.

2. Click Add Level; choose Contact Time and Smallest to Largest.

3. Click OK. The secondary level sorts only within each agent group.

Contact ID	Agent	Time
C207	Harvey	9:30 AM
C210	Harvey	1:00 PM

Contact ID	Agent	Time
C204	Harvey	3:30 PM
C206	Mike Ross	10:00 AM
C209	Mike Ross	11:45 AM
C201	Mike Ross	4:30 PM
C203	Rachel Zane	2:45 PM

14.5 Descending time order

Keep Agent Handled → A to Z, but change Contact Time to Largest to Smallest. This displays later calls first within each person's group.

Agent	Contact ID sequence
Harvey	C204 (3:30 PM), C210 (1:00 PM), C207 (9:30 AM)
Mike Ross	C201 (4:30 PM), C209 (11:45 AM), C206 (10:00 AM)
Rachel Zane	C203 (2:45 PM)

14.6 Three-level sort and dialog controls

Priority	Column	Order
1	Contact Date	Oldest to Newest
2	Agent Handled	A to Z
3	Contact Time	Smallest to Largest

The dialog also provides Add Level, Delete Level, Copy Level, and Move Up/Down. Changing the level order changes the priority. Sort On can use cell values, cell colours, font colours, or conditional-format icons, though these exercises use Cell Values.

LESSON 15 | Advanced Filtering

15.1 Advanced Filter versus AutoFilter

Advanced Filter uses a separately defined criteria range to filter source records in place or copy matching records to another location. Unlike the standard AutoFilter, it does not add dropdown arrows to the headers. Criteria changes are not automatically recalculated; rerun Advanced Filter after editing the cells.

Field in dialog	Meaning
Filter the list, in-place	Hide nonmatching rows in the source area
Copy to another location	Output matching records to a destination on the active sheet
List range	Source table, including headers
Criteria range	Matching header(s) and conditions in separate cells
Copy to	Top-left output destination, when copying
Unique records only	Return distinct selected output records

15.2 Practice dataset: contact transactions

ID	Contact Date	Agent	Type	Duration	Rating	Product	Price
C101	02-Jan-23	Mike Ross	Query	180	8	Crispy Pizza	150
C102	05-Jan-23	Harvey	Complaint	260	5	Cheese Pizza	200
C103	07-Jan-23	Rachel	Request	120	9	Veg Burger	90
C104	10-Jan-23	Mike Ross	Query	200	7	Crispy Burger	120
C105	12-Jan-23	Harvey	Complaint	310	4	Cheese Pizza	180
C106	15-Jan-23	Mike Ross	Query	150	8	Crispy Pizza	150
C107	18-Jan-23	Rachel	Request	90	9	Veg Burger	90
C108	20-Jan-23	Harvey	Complaint	280	6	Crispy Pizza	170
C109	24-Jan-23	Mike Ross	Query	175	8	Veg Pizza	110
C110	28-Jan-23	Harvey	Complaint	260	7	Crispy Burger	95
C111	30-Jan-23	Mike Ross	Request	350	5	Veg Burger	80
C112	31-Jan-23	Harvey	Complaint	420	3	Crispy Pizza	90

Important: The worksheet headers for the eight-column dataset are exactly: Contact ID, Contact Date, Agent Handled, Contact Type, Conversation Duration, Rating Given, Product Name, Product Price. Use these exact full labels in criteria headers.

15.3 Single criterion: Rating Given >7

Keep the data in A1:H13. Set J16 = “Rating Given” and J17 = “>7”. Placing criteria below row 13 avoids having the criteria row hidden by an in-place filter.

1. Select A1:H13. Choose Data → Sort & Filter → Advanced.

2. Select Filter the list, in-place.

3. List range = $A$1:$H$13; Criteria range = $J$16:$J$17; Unique records only unchecked.

4. Click OK.

Matching Contact IDs	Ratings
C101, C103, C106, C107, C109	8, 9, 8, 9, 8

Change J17 to “>8”, clear the previous filter, and reapply Advanced Filter to get only C103 and C107 (both rating 9). Editing the criterion alone does not refresh the displayed rows.

15.4 Criteria operators and literal values

Criterion	Meaning
>7	Greater than 7
<6	Less than 6
>=7	At least 7
<=100	At most 100
260	Numeric value equals 260
<>Complaint	Not equal to Complaint

Excel Advanced Filter treats conditions on the same criteria row as AND; different criteria rows are OR groups. The criteria range must contain the correct header row and each condition row.

15.5 Multiple criteria in the same row (AND)

Put these criteria below the source table, for example J20:L21:

Conversation Duration	Contact Type	Rating Given
>150	Complaint	<6

1. Clear any existing filter; select A1:H13.

2. Data → Advanced → Filter the list, in-place.

3. Set Criteria range to the three-column header-and-value range J20:L21, then OK.

Matches	Why
C102 (260 s, rating 5)	Complaint; duration >150; rating <6
C105 (310 s, rating 4)	All three conditions pass
C112 (420 s, rating 3)	All three conditions pass

15.6 OR conditions in different criteria rows

To show Contact Type = Complaint OR Rating Given >8, use a two-column, three-row criteria block (e.g. J25:K27):

Contact Type	Rating Given
Complaint	[leave blank]
[leave blank]	>8

Result: C102, C103, C105, C107, C108, C110, C112 (7 records). Rows C103 and C107 qualify through their rating even though their type is Request.

15.7 Combining AND groups using OR

Contact Type	Rating Given
Complaint	<6
Query	>7

This means (Complaint AND rating <6) OR (Query AND rating >7). Expected IDs: C101, C102, C105, C106, C109, C112 (6).

15.8 Exact matching on multiple fields

Conversation Duration	Contact Type	Rating Given
260	Complaint	7

Expected match: C110 only. C102 shares the same duration and type but has rating 5, so it is excluded. Numerical criteria written as plain 260 match the value 260.

15.9 Copy matching rows to another area

The screenshot-guided exercise used Contact Type = Complaint AND Conversation Duration >150. The user's settings were List range A1:H13 and Criteria range E20:F21. The crucial change in the Advanced Filter dialog is to select Copy to another location rather than Filter the list, in-place.

Setting	Value
Action	Copy to another location
List range	$A$1:$H$13
Criteria range	$E$20:$F$21
Copy to	$A$25
Unique records only	Unchecked

Expected copied matching IDs: C102, C105, C108, C110, C112 (5 records). Copying generates a separate table starting at A25 while the original data remains available. The earlier separate example "Request AND duration >300" returns C111 only.

15.10 Copy matching rows to another worksheet

The transcript demonstrates why Excel may show "You can only copy filtered data to the active sheet." To output on a separate sheet, start Advanced Filter from the destination worksheet.

1. Create/activate AdvancedFilter_Output, then select the intended output area, e.g. A1.

2. Enter criteria on the output sheet, e.g. J1:M2 with headers Conversation Duration, Contact Type, Agent Handled, Product Price and values >300, Request, Mike Ross, <100.

3. While the output sheet remains active, choose Data → Advanced and select Copy to another location.

4. Use List range from 15_AdvancedFiltering!$A$1:$H$13; Criteria range from AdvancedFilter_Output!$J$1:$M$2; Copy to $A$1 on the active output sheet.

5. Click OK. The output is a separate snapshot (not a live formula or automatic refresh).

Expected output ID	Type	Duration	Agent	Price
C111	Request	350	Mike Ross	80

> Important: Using a full-column list range can accidentally include extra content and blank rows. For this practical dataset the explicit source range A1:H13 is safer. Keep criteria outside any output-overwrite area.

15.11 Unique records only

When checked, this option returns distinct records across the columns included in the output. It is not the same as deleting duplicates from the original data.

Customer	Product
Mike	Pizza
Harvey	Burger
Mike	Pizza
Rachel	Fries
Harvey	Burger
Mike	Burger

1. Select source range A1:B7 (headers included).

2. Data → Advanced → Copy to another location.

3. Leave Criteria range blank; set Copy to $D$1; check Unique records only; click OK.

Customer	Product
Mike	Pizza
Harvey	Burger
Rachel	Fries
Mike	Burger

The two source pairs “Mike / Pizza” and “Harvey / Burger” each appeared twice; only one of each is copied. “Mike / Burger” remains because its Product is different. If you instead select only A1:A7 and copy unique values to F1, the output is Mike, Harvey, Rachel (3 customers).

15.12 Advanced Filter troubleshooting and limitations

Issue	Resolution / explanation
All rows remain visible	Confirm you used Filter the list, in-place, correct list range, criterion text and criterion headers; press OK to apply.
Criteria vanished after in-place filter	Rows are hidden across the sheet; place criteria below the data or on another sheet.
Criteria change does not update result	Clear and rerun Advanced Filter.
Copy-to-sheet error	Activate the destination sheet before opening Advanced Filter.
Unexpected empty result	Verify all AND conditions and actual date/number types.
Normal filter dropdowns not visible	Advanced Filter works without adding normal dropdown arrows.

REFERENCE | Key commands and takeaways

Command / principle	Meaning
Ctrl+A inside a connected dataset	Select current data region
Ctrl+Shift+L	Toggle normal Filter
Home → Sort & Filter	Basic sorting and filtering actions
Data → Clear	Clear current filter restrictions
Data → Sort	Custom multi-level sorting
Data → Advanced	Advanced Filter dialog
Same criteria row	AND conditions
Different criteria rows	OR conditions
Criteria header matches source header	Required to target the correct field
Advanced Filter copied results	Static extraction; rerun after updates

End of Part 3. This document consolidates the completed Learning Sessions 12–15 and excludes assignments and quizzes, as requested.